

Waves

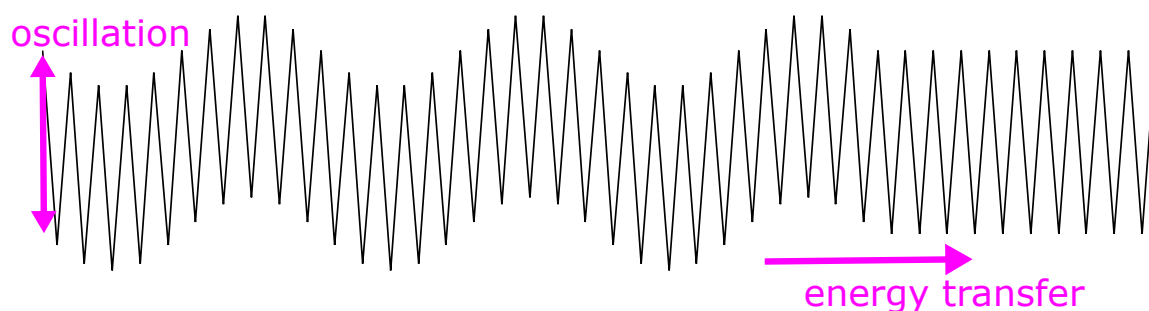
A wave carries **energy** and can carry **information** without carrying **material** from one place to another.

Inside a wave, each part moves one way then the other. This is called **oscillation**. Each part's oscillation causes the next part to oscillate as well.

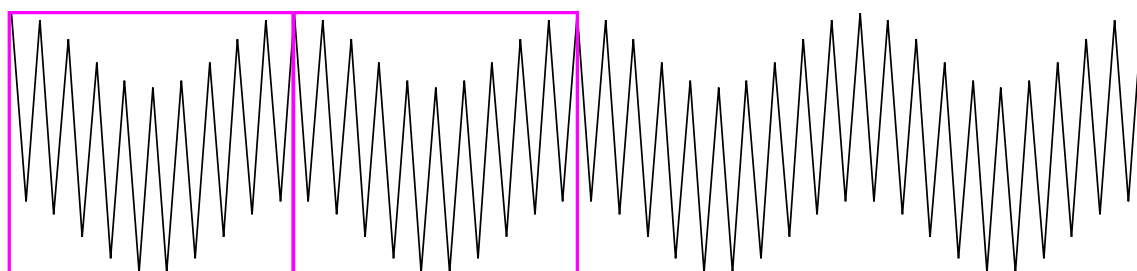
In a **longitudinal** wave the oscillations are in the **same direction** as the energy transfer.

In a **transverse** wave the oscillations are at **right angles** to the energy transfer.

- 1 Two friends stretch a long spring between them. One of them starts moving it up and down. After a short time, it looks like this:



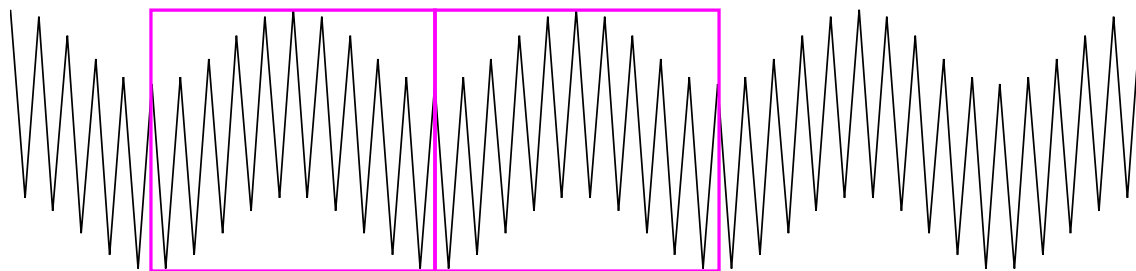
- (a) Is this a transverse or longitudinal wave?
- (b) The highest points on the wave are called **peaks**. Label 3 peaks with the letter P.
- (c) The lowest points are called **troughs**. Label 3 troughs with the letter T.
- 2 Use a ruler to measure the distance between neighbouring peaks in question 1.
- 3 Use a ruler to measure the distance between neighbouring troughs in question 1.
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- 4 A short time later, the wave looks like this. Boxes have been drawn around sections of the wave.



- (a) Compare the contents of each box. Do they look the same?
- (b) Each box contains a section of wave from one _____ to the next.
- (c) Measure the length of each box.

(d) Draw the next box on the wave above.

5 The boxes can be drawn differently on the same wave.



(a) Each box contains a section of wave from one _____ to the next.

(b) Measure the length of each box.

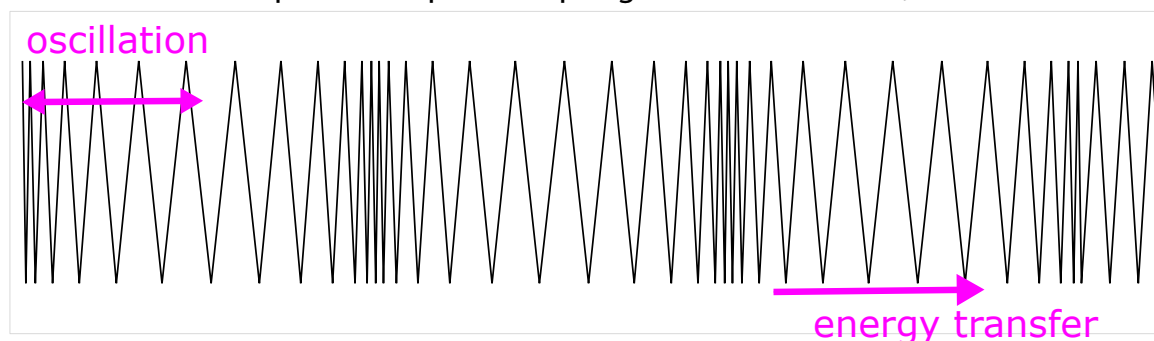
(c) Draw the next box on the wave above.

6 The boxes in questions 4 and 5 show a **repeating unit** in the wave. Their width is called the **wavelength** of the wave.

(a) Write down the wavelength of the wave.

(b) Compare your answer to (a) with your answers to questions 2 and 3.

7 The two friends now push then pull the spring. After a short time, it looks like this:



(a) Is this a transverse or longitudinal wave?

(b) The places where the spring is bunched up on the wave are called **compressions**. Label 3 compressions with the letter C.

(c) The places where the spring is most spaced out are called **rarefactions**. Label 3 rarefactions with the letter R.

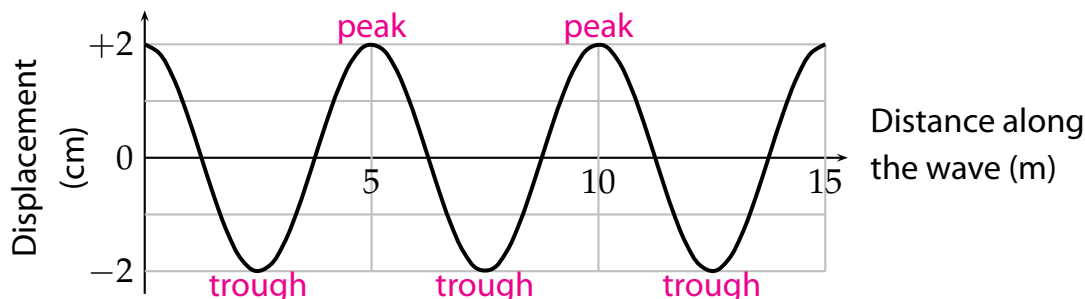
(d) Draw a box around a repeating unit on the wave.

8 Measure the wavelength of the wave in question 7.

Displacement measures how far each part of the spring is from its resting place.

When we draw a graph of a wave, we plot

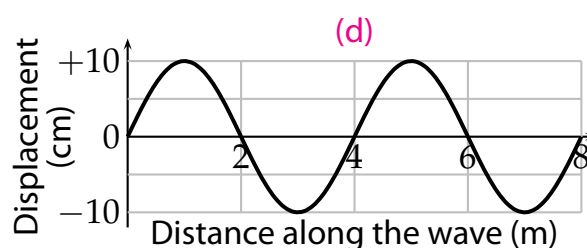
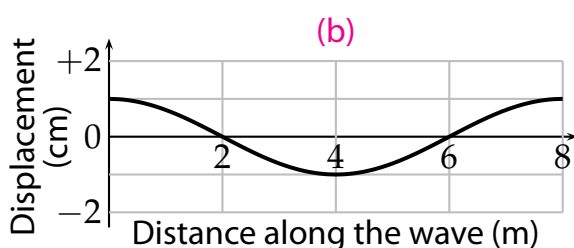
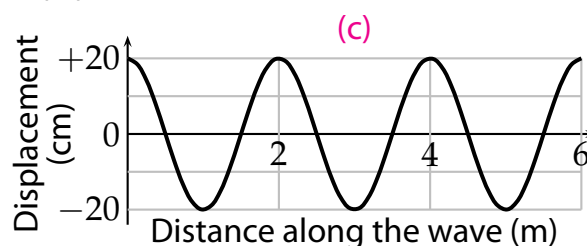
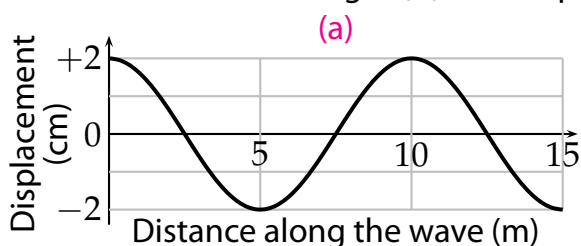
- the **displacement** on the vertical (y) axis, and
- the **distance along the wave** on the horizontal (x) axis.



The **amplitude** (A) is the largest displacement on the wave. Here, the amplitude is **2 cm**.

9 Using the scale on the graph, write down the wavelength of the wave above.

10 Write down the wavelength (λ) and amplitude (A) of each wave below.



Sound is an example of a **longitudinal** wave.

11 A citizen listens to the news on their TV. Complete the sentences using the words or phrases **towards**, **ears**, **brain**, **sound wave**, **speaker**, **oscillates**.

The _____ in the TV _____.

This makes air molecules move away from the speaker then _____ it.

The air carries the _____ to the listener.

When the wave reaches the listener, the air makes parts of their _____ oscillate.

Nerves detect this motion, and the information reaches the listener's _____.